A Technical Note on the $^3P_0$ Model

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1 Introduction

Several excited charmed baryons were first observed by the BaBar and Belle collaborations [1–7]. These discoveries subsequently stimulated extensive theoretical studies of their strong decay properties within the framework of the $^3P_0$ quark model.

Over the past several decades, this model has remained one of the most widely used approaches for investigating charmed and bottom baryons [8–12]. Numerous studies have further adopted the same framework to explore hadronic strong decays in greater detail. For instance, Ref. [13] analyzed the strong decays of the S-wave and P-wave charmed baryons, including $\Sigma_c(2455)$, $\Sigma_c^*(2520)$, and $\Xi_c^*(2645)$, and compared the results with the available experimental data. The same set of parameters was later extended to the P-wave states $\Lambda_c^+(2593)$, $\Lambda_c^+(2625)$, $\Xi_c^{+,0}(2790)$, and $\Xi_c^{+,0}(2815)$, paving the way for further developments of the $^3P_0$ quark model.

The $\Lambda_c(2940)$ was later investigated as a candidate for the $2P$ excited state within this approach [14]. In that work, it was assigned as either $\Lambda_{c1}\left(\frac{1}{2}^-, 2P\right)$ or $\Lambda_{c1}\left(\frac{3}{2}^-, 2P\right)$, with the results indicating that the $J^P = \frac{3}{2}^-$ assignment is more plausible. This study also refined the harmonic-oscillator parameters of mesons, allowing the theoretical predictions to achieve closer agreement with the experimental data.

Along the same line, the Ξ_c candidates were systematically studied as possible P-wave and D-wave excited states within the $^3P_0$ quark model [8, 15]. The analysis was subsequently extended to charmed strange baryons with possible $2S$- and $1D$-excitation assignments for $\Xi_c(2930)^+$, $\Xi_c(2980)^+$, $\Xi_c(3055)^+$, $\Xi_c(3080)^+$, and $\Xi_c(3123)^+$ [16], as well as to the newly observed $\Sigma_b(6097)^\pm$ baryons. Although these studies still contain certain uncertainties in the spectroscopic assignments, they provide an important foundation for improving the model parameters, particularly in distinguishing the S-, P-, D-, and F-wave orbital configurations of baryons. To further clarify the nature of these excited baryon states, more precise experimental measurements of their existence, masses, and total decay widths remain necessary.

In this work, we systematically investigate the strong decay widths of both charmed and bottom baryons within the framework of the $^3P_0$ quark model. Starting from the established parameter set in early studies [13], we perform a detailed comparison with experimental data to validate the numerical framework.

We further investigate the sensitivity of the model predictions to key parameters, including the meson size parameter R, the baryon width parameters β_ρ and β_λ, and the quark-pair creation strength γ. Based on this study, an updated parameter set is proposed to achieve better agreement with recent high-precision experimental data.

This study aims not only to provide a consistent description of $1S$ and $1P$ heavy baryons, but also to offer a refined theoretical baseline for future investigations of excited charmed and bottom baryon states.

2 What is the $^3P_0$ quark model ?

The $^3P_0$ model, also known as the **Quark Pair Creation (QPC)** model, is a phenomenological framework widely used to describe **OZI-allowed** hadronic decays. Originally proposed by Micu [17] and further developed by the Orsay group [18], it has been successfully applied

to explain baryon decays. In this model, a quark-antiquark pair ($q\bar{q}$) is created from the vacuum with $J^{PC} = 0^{++}$, which corresponds to the spectroscopic state ${}^{2S+1}L_J = {}^3P_0$ (where $S = 1$, $L = 1$, and $J = 0$).

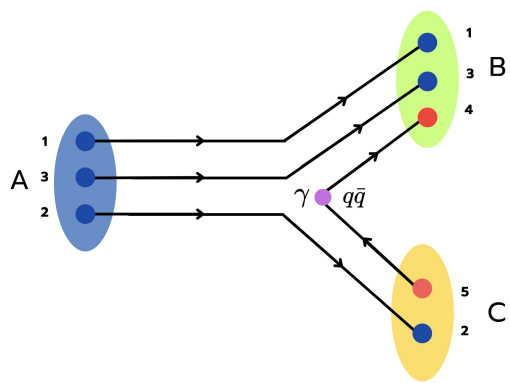

Figure 1: The decay process of $A \to B + C$ in ${}^3P_0$ model.

The decay process, as illustrated in Fig. 1, occurs when this newly created $q\bar{q}$ pair interacts with the constituent quarks of the initial baryon A. For baryon decays, the created pair combines with the original qqq constituents and reorganizes into the outgoing meson–baryon final state through a quark rearrangement process.

3 Background on the Harmonic Oscillator

To evaluate the strong decay amplitudes in the ${}^3P_0$ framework, the orbital wave functions of the heavy baryons must be specified. In this work, we adopt the simple harmonic oscillator (SHO) basis, where the three-quark dynamics are separated into the center-of-mass motion and two internal Jacobi modes, namely the ρ- and λ-coordinates.

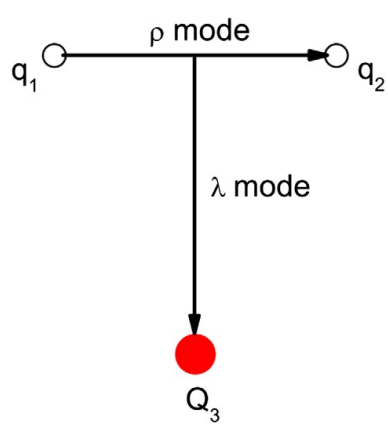

Figure 2: Jacobi coordinates $\vec{\lambda}$ and $\vec{\rho}$ for the heavy baryon system.

The harmonic-oscillator Hamiltonian describing the orbital motion of the three-quark system is written as [19]

$$H = -\sum_{i=1}^{3} \frac{\vec{\nabla}_i^2}{2m_i} + \sum_{i\neq j} \frac{k}{2} \left(\vec{r}_i - \vec{r}_j\right)^2 , \tag{1}$$

where the first term is the kinetic energy of the constituent quarks, and the second term is an effective quadratic confining interaction.

For two light quarks q_1, q_2 with equal masses m and one charm quark Q_3 with mass M, the Hamiltonian can be separated into

$$\begin{aligned} H &= H_G + H_\rho + H_\lambda, \\ H_G &= -\frac{\nabla_X^2}{2(2m+M)}, \\ H_\rho &= -\frac{\nabla_\rho^2}{2m_\rho} + \frac{m_\rho \omega_\rho^2}{2} \vec{\rho}^{\,2}, \\ H_\lambda &= -\frac{\nabla_\lambda^2}{2m_\lambda} + \frac{m_\lambda \omega_\lambda^2}{2} \vec{\lambda}^{\,2}, \end{aligned} \tag{2}$$

where the ρ-mode describes the relative motion of the two light quarks, while the λ-mode corresponds to the motion of the heavy quark relative to the light-quark subsystem.

The Jacobi coordinates are defined as

$$\vec{\rho} = \vec{r}_1 - \vec{r}_2, \tag{3}$$

$$\vec{\lambda} = \frac{1}{2}\left(\vec{r}_1 + \vec{r}_2\right) - \vec{r}_3, \tag{4}$$

with reduced masses

$$m_\rho = \frac{m}{2}, \qquad m_\lambda = \frac{2mM}{2m+M}, \tag{5}$$

and oscillator frequencies

$$\omega_\rho = \sqrt{\frac{3k}{m}}, \qquad \omega_\lambda = \sqrt{\frac{k(2m+M)}{mM}}. \tag{6}$$

The total orbital wave function factorizes as

$$\Psi(\vec{r}_1, \vec{r}_2, \vec{r}_3) = \psi_\lambda(\vec{\lambda})\psi_\rho(\vec{\rho}) e^{i\vec{P}\cdot\vec{X}}, \tag{7}$$

which separates the internal motion from the center-of-mass motion.

The SHO basis is particularly convenient because it has simple analytic forms in both coordinate and momentum spaces [20]:

$$\Psi_{nLM_L}(\mathbf{r}) = R_{nL}(r,\beta)\, Y_{LM_L}(\Omega_r), \tag{8}$$
$$\Psi_{nLM_L}(\mathbf{p}) = R_{nL}(p,\beta)\, Y_{LM_L}(\Omega_p). \tag{9}$$

Here, n, L, and M_L denote the radial, orbital, and magnetic quantum numbers, respectively, while β determines the spatial width of the wave function.

The radial wave functions are

$$R_{nL}(r,\beta) = \beta^{3/2} N_{nL} (\beta r)^L e^{-\beta^2 r^2/2} L_n^{L+1/2}(\beta^2 r^2), \tag{10}$$
$$R_{nL}(p,\beta) = \frac{(-1)^n (-i)^L}{\beta^{3/2}} N_{nL} e^{-p^2/(2\beta^2)} \left(\frac{p}{\beta}\right)^L L_n^{L+1/2}\left(\frac{p^2}{\beta^2}\right), \tag{11}$$

with normalization constant

$$N_{nL} = \sqrt{\frac{2n!}{\Gamma(n+L+3/2)}}. \tag{12}$$

The solid harmonic polynomial is related to the spherical harmonic through

$$y_{LM}(\vec{p}) = |\vec{p}|^L Y_{LM_L}(\Omega_p), \tag{13}$$

which frequently appears in the $^3P_0$ transition operator to represent the momentum-space P-wave structure of the created quark–antiquark pair.

3.1 Radial Functions $R_{nL}(r)$ in Coordinate Space for SPDF Orbitals

$$
\begin{aligned}
1S: \quad & R_{00}(r) = \frac{2a^{3/2}e^{-\frac{1}{2}a^2r^2}}{\pi^{1/4}}, \\
1P: \quad & R_{01}(r) = \frac{2\sqrt{\frac{2}{3}}a^{5/2}re^{-\frac{1}{2}a^2r^2}}{\pi^{1/4}}, \\
1D: \quad & R_{02}(r) = \frac{4a^{7/2}r^2e^{-\frac{1}{2}a^2r^2}}{\sqrt{15}\pi^{1/4}}, \\
1F: \quad & R_{03}(r) = \frac{4\sqrt{\frac{2}{105}}a^{9/2}r^3e^{-\frac{1}{2}a^2r^2}}{\pi^{1/4}}, \\
2S: \quad & R_{10}(r) = \frac{2\sqrt{\frac{2}{3}}a^{3/2}\left(\frac{3}{2}-a^2r^2\right)e^{-\frac{1}{2}a^2r^2}}{\pi^{1/4}}, \\
2P: \quad & R_{11}(r) = \frac{4a^{5/2}r\left(\frac{5}{2}-a^2r^2\right)e^{-\frac{1}{2}a^2r^2}}{\sqrt{15}\pi^{1/4}}, \\
2D: \quad & R_{12}(r) = \frac{4\sqrt{\frac{2}{105}}a^{7/2}r^2\left(\frac{7}{2}-a^2r^2\right)e^{-\frac{1}{2}a^2r^2}}{\pi^{1/4}} \\
2F: \quad & R_{13}(r) = \frac{8a^{9/2}r^3\left(\frac{9}{2}-a^2r^2\right)e^{-\frac{1}{2}a^2r^2}}{3\sqrt{105}\pi^{1/4}}
\end{aligned}
\tag{14}
$$

The radial profile of the ground-state wave function $R_{nL}(r)$ is displayed below. The corresponding excited-state profile is also presented for comparison.

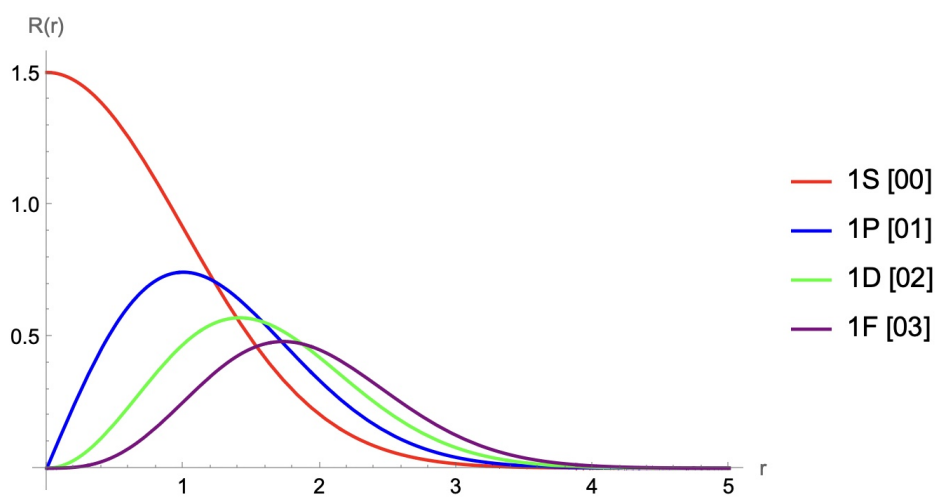

Figure 3:

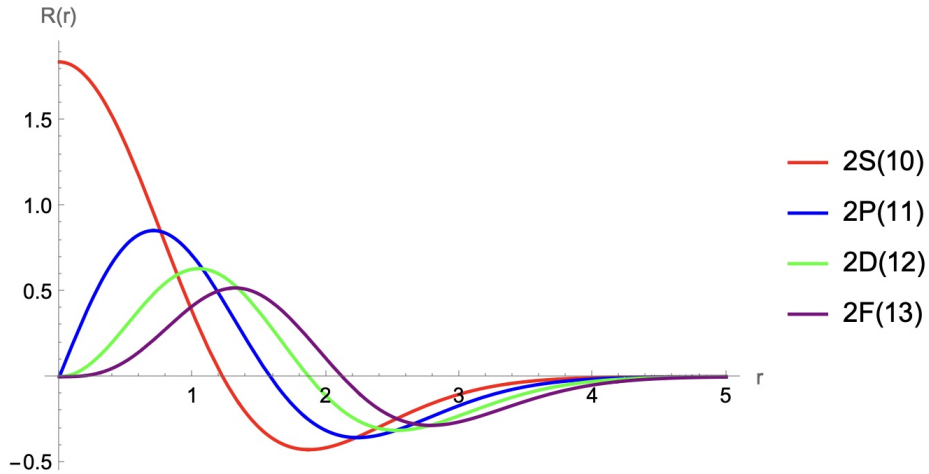

Figure 4:

3.2 Radial Function $R_{nL}(p)$ in Momentum-Space Wave Functions for SPDF orbitals

$$
\begin{aligned}
1S: \quad & R_{00}(p) = \frac{2e^{-\frac{p^2}{2a^2}}}{\pi^{\frac{1}{4}} a^{\frac{3}{2}}} \\
1P: \quad & R_{01}(p) = -\frac{2\sqrt{\frac{2}{3}} ipe^{-\frac{p^2}{2a^2}}}{\pi^{\frac{1}{4}} a^{\frac{5}{2}}} \\
1D: \quad & R_{02}(p) = \frac{4i^2p^2e^{-\frac{p^2}{2a^2}}}{\sqrt{15}\pi^{\frac{1}{4}} a^{\frac{7}{2}}} \\
1F: \quad & R_{03}(p) = -\frac{4\sqrt{\frac{2}{105}} i^3p^3e^{-\frac{p^2}{2a^2}}}{\pi^{\frac{1}{4}} a^{\frac{9}{2}}} \\
2S: \quad & R_{10}(p) = -\frac{2\sqrt{\frac{2}{3}} e^{-\frac{p^2}{2a^2}} \left(\frac{3}{2} - \frac{p^2}{a^2}\right)}{\pi^{\frac{1}{4}} a^{\frac{3}{2}}} \\
2P: \quad & R_{11}(p) = \frac{4ipe^{-\frac{p^2}{2a^2}} \left(\frac{5}{2} - \frac{p^2}{a^2}\right)}{\sqrt{15}\pi^{\frac{1}{4}} a^{\frac{5}{2}}} \\
2D: \quad & R_{12}(p) = -\frac{4\sqrt{\frac{2}{105}} i^2p^2e^{-\frac{p^2}{2a^2}} \left(\frac{7}{2} - \frac{p^2}{a^2}\right)}{\pi^{\frac{1}{4}} a^{\frac{7}{2}}} \\
2F: \quad & R_{13}(p) = \frac{8i^3p^3e^{-\frac{p^2}{2a^2}} \left(\frac{9}{2} - \frac{p^2}{a^2}\right)}{3\sqrt{105}\pi^{\frac{1}{4}} a^{\frac{9}{2}}}
\end{aligned}
\tag{15}
$$

The momentum-space radial profile of the ground-state wave function $R_{nL}(p)$ is shown below in terms of its real and imaginary parts as functions of p, together with its magnitude $|R_{nL}(p)|$. The corresponding excited-state profile is also presented for comparison.

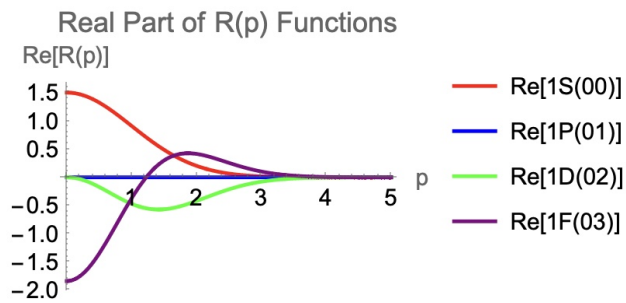

Figure 5:

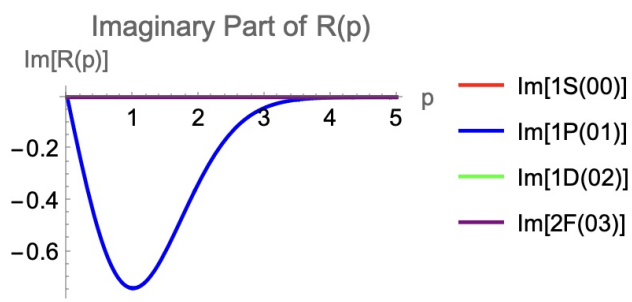

Figure 6:

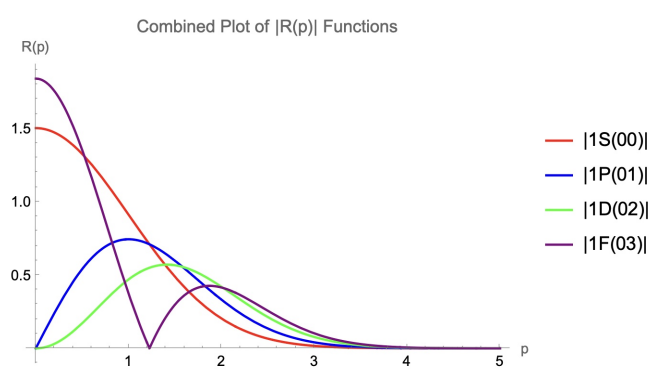

Figure 7:

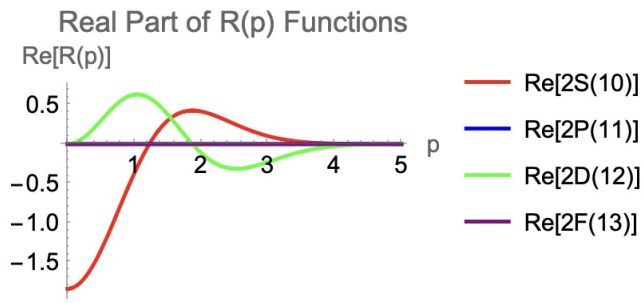

Figure 8:

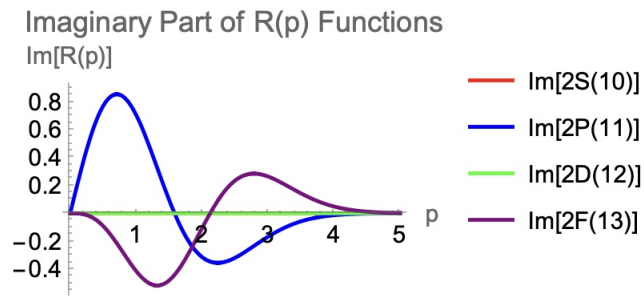

Figure 9:

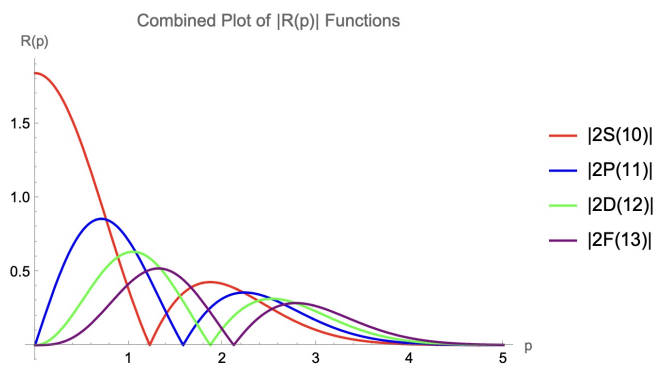

Figure 10:

4 Transition Operator

In the $^3P_0$ model, the hadronic decay process $A \to BC$ proceeds through the creation of a light $q\bar{q}$ pair from the vacuum with quantum numbers $J^{PC} = 0^{++}$. In the nonrelativistic limit, the corresponding transition operator can be expressed as [13]

$$T = -3\gamma \sum_m \langle 1m; 1{-}m|00\rangle \int d^3\mathbf{p}_4\, d^3\mathbf{p}_5\, \delta^3(\mathbf{p}_4 + \mathbf{p}_5) \mathcal{Y}_1^m\left(\frac{\mathbf{p}_4 - \mathbf{p}_5}{2}\right) \times \chi^{45}_{1,-m}\, \phi_0^{45}\, \omega_0^{45}\, b^\dagger_{4i}(\mathbf{p}_4)\, d^\dagger_{5j}(\mathbf{p}_5), \tag{16}$$

where γ denotes the dimensionless strength of the $q\bar{q}$ pair creation from the vacuum. The Clebsch–Gordan coefficient $\langle 1m; 1{-}m|00\rangle$ couples the orbital angular momentum and spin of the created pair into total angular momentum $J = 0$. The delta function $\delta^3(\mathbf{p}_4 + \mathbf{p}_5)$ ensures momentum conservation of the created quark-antiquark pair, implying that the pair is produced with zero total momentum,

$$\mathbf{p}_{\rm cm} = \mathbf{p}_4 + \mathbf{p}_5 = 0, \tag{17}$$

which is consistent with the vacuum quantum numbers. The relative momentum of the created pair is defined as

$$\mathbf{p}_{\rm rel} = \frac{\mathbf{p}_4 - \mathbf{p}_5}{2}. \tag{18}$$

The solid harmonic polynomial is defined as

$$\mathcal{Y}_{1m}(\mathbf{p}_{\rm rel}) = |\mathbf{p}_{\rm rel}|\, Y_{1m}(\Omega_p), \tag{19}$$

which describes the P-wave orbital motion of the created quark–antiquark pair. The spin wave function $\chi^{45}_{1,-m}$ corresponds to a spin-triplet state with $S = 1$, whereas ϕ_0^{45} and ω_0^{45} denote the flavor-singlet and color-singlet wave functions, respectively.
The flavor-singlet wave function in SU(3) is

$$\phi_0^{45} = \frac{1}{\sqrt{3}}\left(u\bar{u} + d\bar{d} + s\bar{s}\right), \tag{20}$$

while in SU(2) it reduces to

$$\phi_0^{45} = \frac{1}{\sqrt{2}}\left(u\bar{u} + d\bar{d}\right). \tag{21}$$

The color-singlet wave function is

$$\omega_0^{45} = \delta_{ij}, \tag{22}$$

which guarantees color conservation in the pair-creation process. The overall factor of 3 arises from summing over the three color degrees of freedom.

4.1 Mock State

The hadronic states appearing in the $^3P_0$ model are described using the mock-state formalism [21]. For a baryon composed of three constituent quarks, the state is written as a direct product of spatial, spin, flavor, and color wave functions, coupled to the total angular momentum.

4.1.1 Initial baryon A

The initial baryon state A is defined as

$$\begin{aligned}\left|A\left(n_A^{2S_A+1}L_{A(J_A,M_{J_A})}\right)(\vec{P}_A)\right\rangle = \sqrt{2E_A}\sum_{M_{L_A},M_{S_A}}&\langle L_A M_{L_A}; S_A M_{S_A} \mid J_A M_{J_A}\rangle \\ &\times \int d^3\vec{p}_1\, d^3\vec{p}_2\, d^3\vec{p}_3\, \delta^3(\vec{p}_1+\vec{p}_2+\vec{p}_3-\vec{P}_A)\Psi_{n_A L_A M_{L_A}}(\vec{p}_1,\vec{p}_2,\vec{p}_3) \\ &\times \chi^{123}_{S_A M_{S_A}}\,\varphi^{123}_A\,\omega^{123}_A\ |q_1(\vec{p}_1)\,q_2(\vec{p}_2)\,q_3(\vec{p}_3)\rangle\,. \end{aligned} \tag{23}$$

Here, $\vec{p}_i$ denotes the momentum of quark q_i $(i = 1, 2, 3)$, and $\vec{P}_A$ is the total momentum of the baryon. The functions Ψ, χ, φ, and ω represent the spatial, spin, flavor, and color wave functions, respectively.

4.1.2 Final baryon B

The final baryon state B is similarly written as

$$\begin{aligned}\left|B\left(n_B^{2S_B+1}L_{B(J_B,M_{J_B})}\right)(\vec{P}_B)\right\rangle = \sqrt{2E_B}\sum_{M_{L_B},M_{S_B}}&\langle L_B M_{L_B}; S_B M_{S_B} \mid J_B M_{J_B}\rangle \\ &\times \int d^3\vec{p}_1\, d^3\vec{p}_2\, d^3\vec{p}_4\, \delta^3(\vec{p}_1+\vec{p}_2+\vec{p}_4-\vec{P}_B)\Psi_{n_B L_B M_{L_B}}(\vec{p}_1,\vec{p}_2,\vec{p}_4) \\ &\times \chi^{124}_{S_B M_{S_B}}\,\varphi^{124}_B\,\omega^{124}_B\ |q_1(\vec{p}_1)\,q_2(\vec{p}_2)\,q_4(\vec{p}_4)\rangle\,. \end{aligned} \tag{24}$$

4.1.3 Final meson C

The final meson state C is given by

$$\begin{aligned}\left|C\left(n_C^{2S_C+1}L_{C(J_C,M_{J_C})}\right)(\vec{P}_C)\right\rangle = \sqrt{2E_C}\sum_{M_{L_C},M_{S_C}}&\langle L_C M_{L_C}; S_C M_{S_C} \mid J_C M_{J_C}\rangle \\ &\times \int d^3\vec{p}_3\, d^3\vec{p}_5\, \delta^3(\vec{p}_3+\vec{p}_5-\vec{P}_C)\Psi_{n_C L_C M_{L_C}}(\vec{p}_3,\vec{p}_5) \\ &\times \chi^{35}_{S_C M_{S_C}}\,\varphi^{35}_C\,\omega^{35}_C\ |q_3(\vec{p}_3)\,q_5(\vec{p}_5)\rangle\,. \end{aligned} \tag{25}$$

For the strong decay of the charmed baryon A, three possible quark rearrangement channels can contribute:

$$A(q_1,q_2,c_3) + (q_4\bar{q}_5) \to B(q_2,q_4,c_3) + C(q_1,\bar{q}_5), \tag{26}$$

$$A(q_1,q_2,c_3) + (q_4\bar{q}_5) \to B(q_1,q_4,c_3) + C(q_2,\bar{q}_5), \tag{27}$$

$$A(q_1,q_2,c_3) + (q_4\bar{q}_5) \to B(q_1,q_2,q_4) + C(c_3,\bar{q}_5). \tag{28}$$

Note that these satisfy the normalization condition

$$\langle A(\boldsymbol{P}_A) \mid A(\boldsymbol{P}'_A)\rangle = 2E_A\delta^3(\boldsymbol{P}_A - \boldsymbol{P}'_A) \tag{29}$$
$$\langle B(\boldsymbol{P}_B) \mid B(\boldsymbol{P}'_B)\rangle = 2E_B\delta^3(\boldsymbol{P}_B - \boldsymbol{P}'_B) \tag{30}$$
$$\langle C(\boldsymbol{P}_C) \mid C(\boldsymbol{P}'_C)\rangle = 2E_C\delta^3(\boldsymbol{P}_C - \boldsymbol{P}'_C) \tag{31}$$

The baryon has a color wave function expressed as

$$\begin{aligned} |B\rangle &= \frac{1}{\sqrt{6}}\epsilon_{abc}\,|q^a q^b q^c\rangle \\ &= \frac{1}{\sqrt{6}}\left(|rgb\rangle + |gbr\rangle + |brg\rangle - |rbg\rangle - |bgr\rangle - |grb\rangle\right). \end{aligned} \tag{32}$$

and meson

$$\begin{aligned} |M\rangle &= \frac{1}{\sqrt{3}}\sum_i |q^i\bar{q}^i\rangle \\ &= \frac{1}{\sqrt{3}}\left(|r\bar{r}\rangle + |g\bar{g}\rangle + |b\bar{b}\rangle\right). \end{aligned} \tag{33}$$

4.2 Helicity Amplitude

Within the $^3P_0$ quark-pair creation framework, the strong decay process $A \to B + C$ is described by the helicity amplitude, which contains the full spin, orbital, flavor, and spatial structure of the transition [8, 13, 15]. For the quark rearrangement channel considered in this work, the helicity amplitude is written as

$$\begin{aligned}
\mathcal{M}^{M_{J_A} M_{J_B} M_{J_C}} = -2\gamma\sqrt{8E_A E_B E_C} \sum_{M_{\rho_A}, M_{L_A}} \sum_{M_{\rho_B}, M_{L_B}} \sum_{M_{S_1}, M_{S_3}, M_{S_4}, m} \\
\times \left\langle J_{l_A} M_{J_{l_A}} S_3 M_{S_3} \mid J_A M_{J_A} \right\rangle \left\langle L_{\rho_A} M_{L_{\rho_A}} L_{\lambda_A} M_{L_{\lambda_A}} \mid L_A M_{L_A} \right\rangle \\
\times \langle L_A M_{L_A} S_{12} M_{S_{12}} \mid J_{12} M_{J_{12}} \rangle \left\langle S_1 M_{S_1} S_2 M_{S_2} \mid J_{l_A} M_{J_{l_A}} \right\rangle \\
\times \left\langle J_{l_B} M_{J_{l_B}} S_3 M_{S_3} \mid J_B M_{J_B} \right\rangle \left\langle L_{\rho_B} M_{L_{\rho_B}} L_{\lambda_B} M_{L_{\lambda_B}} \mid L_B M_{L_B} \right\rangle \\
\times \langle L_B M_{L_B} S_{14} M_{S_{14}} \mid J_{14} M_{J_{14}} \rangle \left\langle S_1 M_{S_1} S_4 M_{S_4} \mid J_{l_B} M_{J_{l_B}} \right\rangle \\
\times \langle 1m; 1-m \mid 00 \rangle \langle S_4 M_{S_4} S_5 M_{S_5} \mid 1-m \rangle \\
\times \langle L_C M_{L_C} S_C M_{S_C} \mid J_C M_{J_C} \rangle \langle S_2 M_{S_2} S_5 M_{S_5} \mid S_C M_{S_C} \rangle \\
\times \left\langle \varphi_B^{1,4,3} \varphi_C^{2,5} \mid \varphi_A^{1,2,3} \varphi_0^{4,5} \right\rangle I^{M_{L_A}, m}_{M_{L_B}, M_{L_C}}(\vec{p}).
\end{aligned} \tag{34}$$

In Eq. (34), γ is the strength of quark pair creation from vacuum. the Clebsch–Gordan coefficients describe the coupling of the orbital, spin, and total angular momenta of the constituent quarks in hadrons A, B, and C. Specifically, $\langle L_\rho M_{L_\rho}\, L_\lambda M_{L_\lambda} \mid L M_L \rangle$ couples the ρ- and λ-mode orbital angular momenta to form the total orbital angular momentum L, while $\langle S_i M_{S_i}\, S_j M_{S_j} \mid S_{ij} M_{S_{ij}} \rangle$ combines the quark spins into the corresponding spin subsystem. These intermediate orbital and spin quantum numbers are then successively coupled through $\langle L M_L\, S M_S \mid J M_J \rangle$ to construct the total angular momentum of the baryons and the meson. The coefficient $\langle 1m; 1-m \mid 00 \rangle$ represents the spin-triplet quark-pair creation from the vacuum with the quantum numbers of the $^3P_0$ state.

The flavor structure of the decay is contained in the matrix element

$$\left\langle \varphi_B^{1,4,3} \varphi_C^{2,5} \mid \varphi_A^{1,2,3} \varphi_0^{4,5} \right\rangle = f(-1)^{I_{12}+I_M+I_A+I_3} \left[\frac{1}{2}(2I_M+1)(2I_B+1)\right]^{1/2} \left\{ \begin{array}{ccc} I_{12} & I_B & I_4 \\ I_M & I_3 & I_A \end{array} \right\}, \tag{35}$$

where $f = \sqrt{2/3}$ or $-\sqrt{1/3}$ for the created quark pair with isospin 1/2 or 0, respectively. Here, I_A, I_B, and I_M denote the isospins of the initial baryon, final baryon, and final meson, while I_{12}, I_3, and I_4 correspond to the relevant constituent quarks.

The quantity $I^{M_{L_A},m}_{M_{L_B},M_{L_C}}(\vec{p})$ represents the spatial overlap integral, defined as

$$\begin{aligned} I^{M_{L_A},m}_{M_{L_B},M_{L_C}}(\vec{p}) = \int & d\vec{p}_1\, d\vec{p}_2\, d\vec{p}_3\, d\vec{p}_4\, d\vec{p}_5\; \delta^3(\vec{p}_1+\vec{p}_2+\vec{p}_3-\vec{p}_A)\delta^3(\vec{p}_4+\vec{p}_5) \\ & \times \delta^3(\vec{p}_1+\vec{p}_4+\vec{p}_3-\vec{p}_B)\delta^3(\vec{p}_2+\vec{p}_5-\vec{p}_C) \\ & \times \Psi^*_B(\vec{p}_1,\vec{p}_4,\vec{p}_3)\Psi^*_C(\vec{p}_2,\vec{p}_5)\Psi_A(\vec{p}_1,\vec{p}_2,\vec{p}_3) y^m_1\left(\frac{\vec{p}_4-\vec{p}_5}{2}\right). \end{aligned} \tag{36}$$

To evaluate this overlap analytically, the hadron wave functions are expanded in the simple harmonic oscillator (SHO) basis. For the baryons, the spatial wave functions are factorized into the ρ- and λ-mode components:

$$\Psi_A(\vec{p}_A) = N\, \Psi_{n_{\rho_A} L_{\rho_A} M_{L_{\rho_A}}}(\vec{p}_{\rho_A})\Psi_{n_{\lambda_A} L_{\lambda_A} M_{L_{\lambda_A}}}(\vec{p}_{\lambda_A}), \tag{37}$$

$$\Psi_B(\vec{p}_B) = N\, \Psi_{n_{\rho_B} L_{\rho_B} M_{L_{\rho_B}}}(\vec{p}_{\rho_B})\Psi_{n_{\lambda_B} L_{\lambda_B} M_{L_{\lambda_B}}}(\vec{p}_{\lambda_B}), \tag{38}$$

whereas the meson wave function contains only a single relative coordinate,

$$\Psi_C(\vec{p}_C) = N\, \Psi_{n_C L_C M_{L_C}}(\vec{p}_C). \tag{39}$$

Here, N is the normalization constant of the total wave function. Because the meson is a two-body system, it does not involve separate ρ- and λ-mode excitations. The explicit forms of the SHO basis function $\Psi_{nLM_L}(\vec{p})$, the associated Laguerre polynomial $L_n^{L+1/2}(\vec{p}^2/\beta^2)$, and the spherical harmonic $Y_{LM_L}(\Omega_p)$ have been introduced in Section 3.

4.2.1 Baryon SHO Wave Functions

As mentioned in eqs. (37) to (39), the total spatial wave function of the **S**-wave charmed baryon can be expressed in terms of $\Psi(L_\rho, m_\rho, L_\lambda, m_\lambda)$ as

$$\Psi(0,0,0,0) = 3^{3/4}\frac{\exp\left(-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}\right)}{\pi^{3/2}\beta_\rho^{3/2}\beta_\lambda^{3/2}}. \tag{40}$$

For the **P**-wave charmed baryons,

$$\Psi(1,m_\rho,0,0) = -i\,3^{3/4}\frac{2\sqrt{2}\,\exp\left(-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}\right) y_{1,m_\rho}(\vec{p}_\rho)}{\sqrt{3\pi}\,\beta_\rho^{5/2}\beta_\lambda^{3/2}}, \tag{41}$$

$$\Psi(0,0,1,m_\lambda) = -i\,3^{3/4}\frac{2\sqrt{2}\,\exp\left(-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}\right) y_{1,m_\lambda}(\vec{p}_\lambda)}{\sqrt{3\pi}\,\beta_\rho^{3/2}\beta_\lambda^{5/2}}. \tag{42}$$

For the **D-wave** charmed baryons, the relevant spatial wave functions are

$$\Psi(0,0,2,m) = 3^{3/4}\frac{4\,e^{-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}}\,y_{2,m_\lambda}(\vec{p}_\lambda)}{\sqrt{15}\pi\beta_\rho^{3/2}\beta_\lambda^{7/2}}, \tag{43}$$

$$\Psi(2,m,0,0) = 3^{3/4}\frac{4\,e^{-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}}\,y_{2,m_\rho}(\vec{p}_\rho)}{\sqrt{15}\pi\beta_\rho^{7/2}\beta_\lambda^{3/2}}, \tag{44}$$

$$\Psi(1,m_\rho,1,m_\lambda) = 3^{3/4}\frac{8\,e^{-\frac{\vec{p}_\rho^{\,2}}{2\beta_\rho^2}-\frac{\vec{p}_\lambda^{\,2}}{2\beta_\lambda^2}}\,y_{1,m_\rho}(\vec{p}_\rho)y_{1,m_\lambda}(\vec{p}_\lambda)}{3\sqrt{\pi}\beta_\rho^{5/2}\beta_\lambda^{5/2}}. \tag{45}$$

Here, the factor $3^{3/4}$ is the normalization constant of the baryon spatial wave function, as obtained in Section D. The parameter R characterizes the harmonic-oscillator wave function of the meson, while β_ρ and β_λ specify the corresponding width parameters of the baryon wave functions.

4.2.2 Meson SHO Wave Function

For the final-state meson, the SHO wave function is taken as

$$\Psi(0,0,0) = \frac{R^{3/2}\exp\left[-\frac{R^2(\vec{p}_2-\vec{p}_5)^2}{8}\right]}{\pi^{3/4}}. \tag{46}$$

Since the meson is a two-body system, it contains only a single relative coordinate and therefore does not involve separate ρ- and λ-mode excitations. Here,

$$\vec{p}_C = \frac{\vec{p}_2-\vec{p}_5}{2}, \qquad R = \frac{1}{\beta_C}. \tag{47}$$

The solid harmonic polynomial $y_{LM}(\vec{p})$ has been defined previously in Eq. (13), where it is related to the spherical harmonic function $Y_{LM}(\Omega_p)$.

4.3 Hadronic Decay Width

Once the helicity amplitude is obtained, the hadronic decay width for the process $A \to B+C$ can be evaluated within the $^3P_0$ framework through

$$\Gamma = \pi^2\frac{|\vec{p}|}{m_A^2}\frac{1}{2J_A+1}\sum_{M_{J_A},M_{J_B},M_{J_C}}\left|\mathcal{M}^{M_{J_A}M_{J_B}M_{J_C}}\right|^2. \tag{48}$$

Here, m_A and J_A denote the mass and total angular momentum of the initial baryon, respectively, while m_B and m_C are the masses of the final-state hadrons. The quantity $\mathcal{M}^{M_{J_A}M_{J_B}M_{J_C}}$ is the helicity amplitude defined in Eq. (34).

The magnitude of the daughter-hadron momentum in the center-of-mass frame of particle A is

$$|\vec{p}| = \frac{\sqrt{[m_A^2 - (m_B - m_C)^2]\,[m_A^2 - (m_B + m_C)^2]}}{2m_A}. \tag{49}$$

5 Predictions for $1S$ and $1P$ Charmed and Bottom Baryons

In this section, we aim to reproduce the hadronic decay widths of known charmed baryons and compare our results with previously reported values in the literature. This comparison serves as a benchmark for the model parameters adopted in the present calculation. First, we calculate the decay widths for $\Sigma_c^{++,+,0}(2455), \Sigma_c^{*++,+,0}(2520) \to \Lambda_c^+\pi$ and $\Sigma_b^{\pm}, \Sigma_b^{*\pm} \to \Lambda_b^0\pi^{\pm}$ in the $1S$ states, adopting the parameters reported in Ref. [13].

It should be noted that this study employs the parameter R to characterize the harmonic-oscillator wave function for the π meson ($R = 2.1$ GeV^{-1}), while β_ρ and $\beta_\lambda = 0.6$ GeV specify the corresponding width parameters of the baryon wave functions, with $\gamma = 13.4$ representing the strength of quark pair creation from the vacuum.

5.1 I S-wave Charmed and Bottom Baryons

State	Mass (MeV)	State	Mass (MeV)	State	Mass (MeV)
$\pi^{\pm}$	139.570	$\Sigma_c^{++}(2455)$	2453.97	Σ_b^{+}	5807.8
π^0	134.977	$\Sigma_c^{+}(2455)$	2452.90	Σ_b^{-}	5815.2
Λ_c^+	2286.46	$\Sigma_c^{0}(2455)$	2453.75	Σ_b^{*+}	5829.0
Λ_b^0	5620.2	$\Sigma_c^{*++}(2520)$	2518.41	Σ_b^{*-}	5836.4
		$\Sigma_c^{*+}(2520)$	2517.36		
		$\Sigma_c^{*0}(2520)$	2518.48		

Table 1:

	J^P	Channel	Decay Width (MeV)		Previous Exp. width 2006 [22]
			Ref.[13]	Our results	
$\Sigma_c^{++}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^+$	1.24	1.18	2.23 ± 0.30
$\Sigma_c^{+}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^0$	1.40	1.35	< 4.6
$\Sigma_c^{0}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^-$	1.24	1.18	2.2 ± 0.40
$\Sigma_c^{*++}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^+$	11.9	11.84	14.9 ± 1.9
$\Sigma_c^{*+}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^0$	12.1	11.92	< 17
$\Sigma_c^{*0}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^-$	11.9	11.86	16.1 ± 2.1

Table 2: The strong decay widths of S-wave charmed baryons $\Sigma_c^{++,+,0}(2455), \Sigma_c^{*++,+,0}(2520)$ in units of MeV. ($\beta_\rho = \beta_\lambda = 0.6$ GeV, $R = 2.1$ GeV^{-1} and $\gamma = 13.4$)

	J^P	Channel	Decay Width (MeV)		Previous Exp. width 2007 [23]
			Ref.[13]	Our results	
Σ_b^+	$\frac{1}{2}^+$	$\Lambda_b^0\pi^+$	3.5	3.7	~ 8
Σ_b^-	$\frac{1}{2}^+$	$\Lambda_b^0\pi^-$	4.7	5.0	
Σ_b^{*+}	$\frac{3}{2}^+$	$\Lambda_b^0\pi^+$	7.5	7.6	~ 15
Σ_b^{*-}	$\frac{3}{2}^+$	$\Lambda_b^0\pi^-$	9.2	9.5	

Table 3: The strong decay widths of S-wave bottom baryons Σ_b, Σ_b^* in units of MeV. ($\beta_\rho = \beta_\lambda = 0.6$ GeV, $R = 2.1$ GeV^{-1} and $\gamma = 13.4$)

5.2 II P-wave Charmed Baryons

After confirming that the adopted parameters can reasonably reproduce the decay widths of the ground-state ($1S$) baryons, we extend the same analysis to the first orbital excited ($1P$) charmed baryons. In this part, we calculate the strong decay widths of $\Lambda_c^+(2593)$, $\Lambda_c^+(2625)$, $\Xi_c^{+,0}(2790)$, and $\Xi_c^{+,0}(2815)$, and compare our results with previous theoretical studies and available experimental data.

	J^P	Channel	Decay Width (MeV)		Previous Exp. width 2006 [22]
			Ref.[13]	Our results	
$\Lambda_c^+(2593)$	$\frac{1}{2}^-$	$\Sigma_c^{++}\pi^-$	3.4	3.7	$3.6^{+2.0}_{-1.3}$
		$\Sigma_c^+\pi^0$	6.4	7.7	
		$\Sigma_c^0\pi^+$	3.4	3.7	
$\Lambda_c^+(2625)$	$\frac{3}{2}^-$	$\Sigma_c^{++}\pi^-$	1.9×10^{-3}	2.1×10^{-3}	< 0.10
		$\Sigma_c^+\pi^0$	2.6×10^{-3}	3.2×10^{-3}	< 1.9
		$\Sigma_c^0\pi^+$	1.9×10^{-3}	2.1×10^{-3}	< 0.10
$\Xi_c^+(2790)$	$\frac{1}{2}^-$	$\Xi_c^{\prime +}\pi^0$	5.0	5.9	< 15
		$\Xi_c^{\prime 0}\pi^+$	4.9	5.7	
$\Xi_c^0(2790)$	$\frac{1}{2}^-$	$\Xi_c^{\prime +}\pi^-$	5.2	5.9	< 12
		$\Xi_c^{\prime 0}\pi^0$	5.1	6.0	
$\Xi_c^+(2815)$	$\frac{3}{2}^-$	$\Xi_c^{\prime +}\pi^0$	2.7	2.3	< 3.5
		$\Xi_c^{\prime 0}\pi^+$	2.6	2.1	
$\Xi_c^0(2815)$	$\frac{3}{2}^-$	$\Xi_c^{\prime +}\pi^-$	2.7	2.3	< 6.5
		$\Xi_c^{\prime 0}\pi^0$	2.8	2.4	

Table 4: The strong decay widths of P-wave charmed baryons $\Lambda_c^+(2593)$, $\Lambda_c^+(2625)$, $\Xi_c^{+,0}(2790)$ and $\Xi_c^{+,0}(2815)$ in units of MeV. ($\beta_\rho = \beta_\lambda = 0.6$ GeV, $R = 2.1$ GeV^{-1} and $\gamma = 13.4$)

Our numerical results effectively reproduce the values reported in the previous literature, confirming that the chosen parameters $R, \beta_\rho, \beta_\lambda$, and γ remain consistent with the experimental status of the 2006–2007 era. Although these parameters provide a reliable baseline for S-wave agreement, it is worth noting that modern experimental measurements have

significantly improved in accuracy. As modern experimental measurements have achieved significantly higher precision, it is anticipated that these model parameters will need further refinement and adjustment to stay consistent with the latest empirical data. These updated values will be further discussed in the context of recent developments in subsequent sections.

5.3 III P-wave Bottom Baryons

In this section, we investigate the strong decay widths of the $\Sigma_b^{\pm}$ and $\Sigma_b^{*\pm}$ states. We evaluate their transitions to $\Lambda_b^0\pi^{\pm}$ by considering them as either ground-state $1S$ baryons or first orbital $1P$ excitations, following the framework and parameters reported by Yang et al. [24]. The model parameters adopted for this calculation are $\beta_\rho = \beta_\lambda = 0.6$ GeV for the S-wave baryon wave functions and $\beta_\rho = \beta_\lambda = 0.4$ GeV for the P-wave baryon wave functions, with $R = 2.5$ GeV^{-1} for the π meson wave function.

The masses of the mesons and baryons used as input for these decay evaluations are summarized in Table 5.

State	Mass (MeV)	State	Mass (MeV)
$\pi^{\pm}$	139.570	Σ_b^-	5815.64
π^0	134.977	Σ_b^+	5810.55
Λ_b^0	5619.58	Σ_b^{*-}	5834.73
$\Lambda_b^0(5912)$	5912.2	Σ_b^{*+}	5830.28
$\Lambda_b^0(5920)$	5919.58		

Table 5: Masses of the mesons and baryons involved in the decay processes for $\Sigma_b^{\pm}, \Sigma_b^{*\pm} \to \Lambda_b^0\pi^{\pm}$.

Table 6 presents the calculated decay widths for the ground-state $1S$ baryons, $\Sigma_{b1}\left(\frac{1}{2}^+\right)$ and $\Sigma_{b1}\left(\frac{3}{2}^+\right)$, as well as the first orbital excited $1P$ states, $\Sigma_{b0}\left(\frac{1}{2}^-\right)$ and $\Sigma_{b2}\left(\frac{3}{2}^-\right)$. Our results are compared with those from Yang et al. [24] and the available experimental averages from the Particle Data Group (PDG) [25].

	Ref. [24]				Our results				PDG (Average) [25]			
$\Sigma_{bJ_l}(J^P)$	Σ_b^+	Σ_b^-	Σ_b^{*+}	Σ_b^{*-}	Σ_b^+	Σ_b^-	Σ_b^{*+}	Σ_b^{*-}	Σ_b^+	Σ_b^-	Σ_b^{*+}	Σ_b^{*-}
$\Sigma_{b1}\left(\frac{1}{2}^+\right)$	4.42	5.32	8.49	9.64	5.55	5.94	10.29	10.97	5.0 ± 0.5	5.3 ± 0.5	–	–
$\Sigma_{b1}\left(\frac{3}{2}^+\right)$	4.42	5.32	8.49	9.64	5.55	5.94	10.29	10.97	–	–	9.4 ± 0.6	10.4 ± 0.6
$\Sigma_{b0}\left(\frac{1}{2}^-\right)$	64.31	69.16	83.10	87.33	60.63	65.63	83.10	86.26	–	–	–	–
$\Sigma_{b2}\left(\frac{3}{2}^-\right)$	0.01	0.01	0.03	0.03	0.01	0.01	0.03	0.03	–	–	–	–

Table 6: Comparison of strong decay widths of Σ_b baryons (in MeV) between the literature, our current calculation, and PDG average values.

The table shows that our numerical results support the assignments of $\Sigma_b^{\pm}$ and $\Sigma_b^{*\pm}$ as $\Sigma_{b1}\left(\frac{1}{2}^+\right)$ and $\Sigma_{b1}\left(\frac{3}{2}^+\right)$, respectively. These states are highly likely to be the ground-state

S-wave Σ_b baryons, characterized by the absence of internal λ or ρ excitations. Although we adopt the same model parameters as those found in the literature, some deviations occur. These differences arise from the different numerical treatment of the spatial overlap integrals; in this study, we follow the computational approach established in Chen et al. [13]. Nevertheless, our reproduced results show better agreement with the PDG [25].

6 IV Updated Predictions for $1S$ and $1P$ Charmed Baryons

In this section, we employ the flavor matrix element formalism introduced in Ref. [24], together with the spatial overlap integrals derived following the same procedure as established in Chen et al. [13]. The harmonic-oscillator parameters are chosen as $\beta_\rho = 400$ MeV, while β_λ is estimated by

$$\beta_\lambda = \left(\frac{3m_Q}{2m_q + m_Q}\right)\beta_\rho \approx 0.485 \text{ GeV}, \tag{50}$$

where we take $m_q = m_u = m_d = 330$ MeV and $m_Q = m_c = 1700$ MeV. In addition, the meson size parameter and the quark-pair creation strength are fixed as $R = 2.5$ GeV^{-1} and $\gamma = 11.51$, respectively.

With this parameter set, we obtain updated predictions for the hadronic decay widths of the established $1S$- and $1P$-wave charmed baryons and compare them with the available PDG averages and experimental measurements.

1S-wave	J^P	Channel	PDG (Average) [25]	EXP1	EXP2	EXP3	$\Gamma_i/\Gamma_{\text{total}}$	Our results
$\Sigma_c^{++}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^+$	$1.83^{+0.09}_{-0.18}$	$1.84 \pm 0.04^{+0.07}_{-0.20}$ (BELL)	$2.34 \pm 0.13 \pm 0.45$ (CDF)	-	$\sim 100\%$	2.11
$\Sigma_c^{+}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^0$	$2.3 \pm 0.3 \pm 0.3$	-	-	-		2.39
$\Sigma_c^{0}(2455)$	$\frac{1}{2}^+$	$\Lambda_c^+\pi^-$	$1.83^{+0.11}_{-0.19}$	$1.76 \pm 0.04^{0.09}_{0.21}$ (BELL)	$1.65 \pm 0.11 \pm 0.49$ (CDF)	-		2.08
$\Sigma_c^{*++}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^+$	$14.78^{+0.30}_{-0.40}$	$14.77 \pm 0.25^{+0.18}_{-0.30}$ (BELL)	$15.03 \pm 2.12 \pm 1.36$ (CDF)	$14.4^{+1.6}_{-1.5} \pm 1.4$ (CLEO) $17.9^{+3.8}_{-3.3} \pm 4$ (CLE2)	$\sim 100\%$	20.55
$\Sigma_c^{*+}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^0$	$17.2^{+2.3+3.1}_{-2.1-0.7}$ (BELL)	-	-	-		21.15
$\Sigma_c^{*0}(2520)$	$\frac{3}{2}^+$	$\Lambda_c^+\pi^-$	$15.3^{+0.4}_{-0.5}$	$15.41 \pm 0.41^{+0.20}_{-0.32}$ (BELL)	$12.51 \pm 1.82 \pm 1.37$ (CDF)	$16.6^{+1.9}_{-1.7} \pm 1.4$ (CLEO) $13.0^{+3.7}_{-3.0} \pm 4$ (CLE2)		20.58

Table 7: Comparison of strong decay widths of $\Sigma_c^{++,+,0}(2455), \Sigma_c^{*++,+,0}(2520)$ baryons (in MeV) between the literature, our current calculation, and PDG average values.

1P-wave	J^P	Channel	PDG (Average) [25]	EXP1	EXP2	$\Gamma_i/\Gamma_{\rm total}$	$\sim \Gamma_{i\rm exp}$ $\left[\frac{\Gamma_i/\Gamma_{\rm total}}{100} \times \Gamma_{\rm PDG}\right]$	Our results
$\Lambda_c(2595)$	$\frac{1}{2}^-$	$\Sigma_c^{++}\pi^-$	$2.59 \pm 0.30 \pm 0.47$ (AALTONEN)	$2.9^{+2.9+1.8}_{-2.1-1.4}$ (ARG)	$3.9^{+1.4+2.0}_{-1.2-1.0}$ (CLE2)	$(24 \pm 7)\%$	~ 0.6216	0.59
		$\Sigma_c^{+}\pi^0$						0.61
		$\Sigma_c^{0}\pi^+$						0.59
$\Lambda_c(2625)$	$\frac{3}{2}^-$	$\Sigma_c^{++}\pi^-$	< 0.52 (WANG)	< 3.2 (ARG)	< 1.9 (CLE2)	$(2.6 \pm 0.4)\%$	~ 0.01352	0.00076
		$\Sigma_c^{+}\pi^0$						0.0011
		$\Sigma_c^{0}\pi^+$						0.00075
$\Xi_c^+(2790)$	$\frac{1}{2}^-$	$\Xi_c^{\prime +}\pi^0$	$8.9 \pm 0.6 \pm 0.8$ (BELL)	–	–	$\approx 100\%$		8.49
		$\Xi_c^{\prime 0}\pi^+$						8.44
$\Xi_c^0(2790)$	$\frac{1}{2}^-$	$\Xi_c^{\prime +}\pi^-$	$10.0 \pm 0.7 \pm 0.8$ (BELL)	–	–	$\approx 100\%$		8.49
		$\Xi_c^{\prime 0}\pi^0$						8.65
$\Xi_c^+(2815)$	$\frac{3}{2}^-$	$\Xi_c^{*+}\pi^0$	$2.43 \pm 0.2 \pm 0.17$ (BELL)	< 3.5 (CLE2)	–			4.36
		$\Xi_c^{*0}\pi^+$						4.09
$\Xi_c^0(2815)$	$\frac{3}{2}^-$	$\Xi_c^{*+}\pi^-$	$2.54 \pm 0.18 \pm 0.17$ (BELL)	< 6.5 (CLE2)	–			4.46
		$\Xi_c^{*0}\pi^0$						4.65

Table 8: Comparison of strong decay widths of $\Lambda_c(2595)$, $\Lambda_c(2625)$, $\Xi_c^{+,0}(2790)$ and $\Xi_c^{+,0}(2815)$ baryons (in MeV) between the literature, our current calculation, and PDG average values.

7 Conclusion

In this study, we have systematically investigated the strong decay widths of $1S$ and $1P$ charmed and bottom baryons within the framework of the $^3P_0$ quark model. By comparing our numerical results with a wide array of experimental data from various eras and previous theoretical studies, we provide a comprehensive assessment of the model's predictive power and the classification of heavy baryon states.

- **Benchmark Validation ($1S$ States):** Initial calculations for the ground-state charmed baryons $\Sigma_c(2455)$, $\Sigma_c^*(2520)$ and bottom baryons Σ_b, Σ_b^* were performed using the parameters $R = 2.1$ GeV^{-1} and $\gamma = 13.4$. Our results successfully reproduced the values reported in early literature [13], validating our numerical treatment of spatial overlap integrals against the experimental status of the 2006–2007 period.

- **Evaluation of $1P$-wave Charmed Baryons:** We extended the analysis to the first orbital excitations, including $\Lambda_c(2593)$, $\Lambda_c(2625)$, $\Xi_c(2790)$, and $\Xi_c(2815)$. While the initial parameters provided a consistent baseline, the comparison with contemporary data highlighted that the model's accuracy is highly sensitive to the meson size parameter R. This comparison served as the basis for further refinement.

- **Bottom Baryon Classification** The investigation of $\Sigma_b^\pm$ and $\Sigma_b^{*\pm}$ transitions to $\Lambda_b^0\pi^\pm$ supports their identification as the S-wave ground states $\Sigma_{b1}(1/2^+)$ and $\Sigma_{b1}(3/2^+)$.

Despite some deviations from previous literature, our calculated widths show improved agreement with the latest PDG 2024 average values [25], particularly through a refined numerical approach for spatial overlap integrals.

- **Updated Predictions** To accommodate the high precision of modern experimental measurements from facilities such as Belle, CDF, and CLEO, we updated our predictions using a refined parameter set ($R = 2.5\ \mathrm{GeV}^{-1}$, $\gamma = 11.51$) and adjusted harmonic-oscillator parameters ($\beta_\rho = 400$ MeV, $\beta_\lambda \approx 485$ MeV). This updated calculation provided a more satisfactory description across the Σ_c and Ξ_c families, effectively narrowing the gap between theoretical predictions and current empirical data.

The $^3P_0$ quark model remains a robust and transparent tool for the classification of heavy baryons. While some decay channels are more satisfactorily described than others, we have demonstrated that by strategically refining the meson size parameter R and the quark-pair creation strength γ, the model can achieve high consistency with experimental observations. These updated results provide a reliable theoretical reference for identifying future orbital and radial excitations of charmed and bottom baryons in current and upcoming collider experiments.

A Coordinate Transformation

Define the relative coordinates as

$$\boldsymbol{\rho} = \boldsymbol{p}_1 - \boldsymbol{p}_2, \quad \boldsymbol{\lambda} = \frac{\boldsymbol{p}_1 + \boldsymbol{p}_2}{2}. \tag{51}$$

The inverse transformation is:

$$\boldsymbol{p}_1 = \boldsymbol{\lambda} + \frac{\boldsymbol{\rho}}{2}, \quad \boldsymbol{p}_2 = \boldsymbol{\lambda} - \frac{\boldsymbol{\rho}}{2}. \tag{52}$$

The Jacobian determinant of this transformation is 1, so $d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 = d^3\boldsymbol{\rho} d^3\boldsymbol{\lambda}$.

B Jacobian Determinant in Spherical Coordinates

Define the spherical coordinates for $\boldsymbol{\rho}$ and $\boldsymbol{\lambda}$:

$$\boldsymbol{\rho} = (\rho, \theta_\rho, \phi_\rho), \quad \boldsymbol{\lambda} = (\lambda, \theta_\lambda, \phi_\lambda). \tag{53}$$

where

$$\rho = |\boldsymbol{\rho}|, \quad \lambda = |\boldsymbol{\lambda}|, \tag{54}$$

and (θ_ρ, ϕ_ρ), $(\theta_\lambda, \phi_\lambda)$ are the respective polar and azimuthal angles.

Since the transformation from $\boldsymbol{p}_1, \boldsymbol{p}_2$ to $\boldsymbol{\rho}, \boldsymbol{\lambda}$ has a Jacobian determinant of 1, the volume element remains unchanged:

$$d^3p_1\, d^3p_2 = d^3\rho\, d^3\lambda. \tag{55}$$

In spherical coordinates, the volume elements are:

$$d^3\rho = \rho^2 d\rho\, d\Omega_\rho, \quad d^3\lambda = \lambda^2 d\lambda\, d\Omega_\lambda, \tag{56}$$

where

$$d\Omega_\rho = d\theta_\rho \sin\theta_\rho d\phi_\rho, \quad d\Omega_\lambda = d\theta_\lambda \sin\theta_\lambda d\phi_\lambda. \tag{57}$$

Thus, the full volume element is:

$$d^3p_1\, d^3p_2 = \rho^2 d\rho\, d\Omega_\rho\, \lambda^2 d\lambda\, d\Omega_\lambda. \tag{58}$$

C Center-of-Mass and Relative Motion (Meson)

the total and relative momenta for a two-particle system with momenta $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ are

$$\text{Total (CM) momentum:} \quad \boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2, \tag{59}$$

$$\text{Relative momentum:} \quad \boldsymbol{p} = \frac{m_2\boldsymbol{p}_1 - m_1\boldsymbol{p}_2}{m_1 + m_2}. \tag{60}$$

The inverse relations are

$$\boldsymbol{p}_1 = \frac{m_1}{m_1 + m_2}\boldsymbol{P} + \boldsymbol{p}, \tag{61}$$

$$\boldsymbol{p}_2 = \frac{m_2}{m_1 + m_2}\boldsymbol{P} - \boldsymbol{p}. \tag{62}$$

The wavefunction naturally factorizes in terms of center-of-mass (CM) and relative motion:

$$\Psi(\boldsymbol{p}_1, \boldsymbol{p}_2) = \Phi_{\mathrm{CM}}(\boldsymbol{P})\psi_{\mathrm{rel}}(\boldsymbol{p}). \tag{63}$$

where $\Phi_{\mathrm{CM}}(\boldsymbol{P})$ describes the motion of the system as a whole and $\psi_{\mathrm{rel}}(\boldsymbol{p})$ describes the internal structure of the bound state.
If the system has a definite total momentum $\boldsymbol{P}_A$, then the CM wavefunction is a plane wave

$$\Phi_{\mathrm{CM}}(\boldsymbol{P}) = \delta^3(\boldsymbol{P} - \boldsymbol{P}_A). \tag{64}$$

The relative motion is described by the quantum numbers of the bound state:

$$\psi_{\mathrm{rel}}(\boldsymbol{p}) = R_{nL}(p)Y_{LM_L}(\hat{p}), \tag{65}$$

If the CM wave function is a plane wave with fixed momentum $\boldsymbol{P}_A$, the wavefunction becomes

$$\Psi(\boldsymbol{p}_1, \boldsymbol{p}_2) = \delta^3(\boldsymbol{P} - \boldsymbol{P}_A)\psi_{\mathrm{rel}}(\boldsymbol{p}). \tag{66}$$

where $\boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2$. The volume element transformation from $d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2$ to $d^3\boldsymbol{P}d^3\boldsymbol{p}$ is given by

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 = |J| d^3\boldsymbol{P} d^3\boldsymbol{p}, \tag{67}$$

where the Jacobian determinant for this transformation matrix J is

$$J = \begin{bmatrix} \frac{m_1}{m_1+m_2}I & I \\ \frac{m_2}{m_1+m_2}I & -I \end{bmatrix}, \tag{68}$$

where I is the 3×3 identity matrix. The determinant is given by:

$$\det J = \left(\frac{m_1}{m_1 + m_2} \cdot (-1) - 1 \cdot \frac{m_2}{m_1 + m_2}\right)^3 = (-1)^3 = 1. \tag{69}$$

Thus, $|\det J| = 1$, implying that

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 = d^3\boldsymbol{P} d^3\boldsymbol{p}. \tag{70}$$

Since the transformation is linear, the volume elements remain equivalent:

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 = d^3\boldsymbol{P} d^3\boldsymbol{p}. \tag{71}$$

D Center-of-Mass and Relative Motion (Baryon)

For a three-particle system with momenta $\boldsymbol{p}_1$, $\boldsymbol{p}_2$, and $\boldsymbol{p}_3$, we define the total and relative momenta as follows

$$\text{Total (CM) momentum:} \quad \boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3, \tag{72}$$

$$\text{Jacobi momenta:} \quad \boldsymbol{p}_\lambda = \sqrt{\frac{1}{6}}(\boldsymbol{p}_1 + \boldsymbol{p}_2 - 2\boldsymbol{p}_3), \tag{73}$$

$$\boldsymbol{p}_\rho = \sqrt{\frac{1}{2}}(\boldsymbol{p}_1 - \boldsymbol{p}_2). \tag{74}$$

The inverse relations for the momenta in terms of the Jacobi momenta $\boldsymbol{p}_\lambda$ and $\boldsymbol{p}_\rho$ are

$$\boldsymbol{p}_1 = \frac{1}{3}\boldsymbol{P} + \frac{1}{\sqrt{6}}\boldsymbol{p}_\lambda + \frac{1}{\sqrt{2}}\boldsymbol{p}_\rho, \tag{75}$$

$$\boldsymbol{p}_2 = \frac{1}{3}\boldsymbol{P} + \frac{1}{\sqrt{6}}\boldsymbol{p}_\lambda - \frac{1}{\sqrt{2}}\boldsymbol{p}_\rho, \tag{76}$$

$$\boldsymbol{p}_3 = \frac{1}{3}\boldsymbol{P} - \sqrt{\frac{2}{3}}\boldsymbol{p}_\lambda. \tag{77}$$

The wavefunction naturally factorizes in terms of center-of-mass (CM) and relative motion:

$$\Psi(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3) = \Phi_{\text{CM}}(\boldsymbol{P})\psi_{\text{rel}}(\boldsymbol{p}_\lambda, \boldsymbol{p}_\rho), \tag{78}$$

where $\Phi_{\text{CM}}(\boldsymbol{P})$ describes the motion of the system as a whole and $\psi_{\text{rel}}(\boldsymbol{p}_\lambda, \boldsymbol{p}_\rho)$ describes the internal structure of the bound state.

If the system has a definite total momentum $\boldsymbol{P}_A$, then the CM wavefunction is a plane wave:

$$\Phi_{\text{CM}}(\boldsymbol{P}) = \delta^3(\boldsymbol{P} - \boldsymbol{P}_A). \tag{79}$$

The relative motion is described by the quantum numbers of the bound state:

$$\psi_{\text{rel}}(\boldsymbol{p}_\lambda, \boldsymbol{p}_\rho) = R_{nL}(p_\lambda, p_\rho)Y_{LM_L}(\hat{p}_\lambda, \hat{p}_\rho), \tag{80}$$

where $R_{nL}(p_\lambda, p_\rho)$ is the radial wavefunction, and $Y_{LM_L}(\hat{p}_\lambda, \hat{p}_\rho)$ are spherical harmonics encoding the angular dependence. If the CM wavefunction is a plane wave with fixed momentum $\boldsymbol{P}_A$, the wavefunction becomes:

$$\Psi(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3) = \delta^3(\boldsymbol{P} - \boldsymbol{P}_A)\psi_{\text{rel}}(\boldsymbol{p}_\lambda, \boldsymbol{p}_\rho). \tag{81}$$

where $\boldsymbol{P} = \boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3$. The volume element transformation from $d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3$ to the center-of-mass and relative momenta $d^3\boldsymbol{P} d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho$ is given by

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 = |\det J|\, d^3\boldsymbol{P} d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho. \tag{82}$$

The transformation matrix J is

$$J = \begin{bmatrix} \frac{1}{3}I & \frac{1}{\sqrt{6}}I & \frac{1}{\sqrt{2}}I \\ \frac{1}{3}I & \frac{1}{\sqrt{6}}I & -\frac{1}{\sqrt{2}}I \\ \frac{1}{3}I & -\sqrt{\frac{2}{3}}I & 0 \end{bmatrix}, \tag{83}$$

where I is the 3×3 identity matrix.

The determinant is given by

$$\det J = \left(\frac{1}{3} \cdot (-\sqrt{\frac{2}{3}}) - \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{2}} \cdot (-\frac{1}{\sqrt{2}}) \right)^3 = \left(-\frac{1}{\sqrt{3}} \right)^3. \tag{84}$$

Taking the absolute value,

$$|\det J| = \left(\frac{1}{\sqrt{3}} \right)^3 = \frac{1}{3^{3/2}}. \tag{85}$$

Thus, the volume element transformation for baryons is:

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 = \frac{1}{3^{3/2}} d^3\boldsymbol{P} d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho. \tag{86}$$

This result accounts for the transformation from individual quark momenta to the center-of-mass and Jacobi momenta in a baryon system.

For the normalization condition, we begin with the definition of the baryon state in terms of the quark wave function in momentum space:

$$|B(\boldsymbol{P}_B)\rangle = \sqrt{2E_B} \int d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 \, \delta^3(\boldsymbol{p}_1+\boldsymbol{p}_2+\boldsymbol{p}_3-\boldsymbol{P}_B) \psi_{n_B L_B M_{L_B}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3) \, |q_1(\boldsymbol{p}_1) q_2(\boldsymbol{p}_2) q_3(\boldsymbol{p}_3)\rangle \, . \tag{87}$$

Here, the wavefunction $\psi_{n_B L_B M_{L_B}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3)$ depends only on $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$, with $\boldsymbol{p}_3$ determined by momentum conservation.

The inner product of two baryon states is:

$$\begin{aligned} \langle B(\boldsymbol{P}_B)\rangle \, B'(\boldsymbol{P}'_B) \;\; = \;\; & 2E_B \int d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 d^3\boldsymbol{p}'_1 d^3\boldsymbol{p}'_2 d^3\boldsymbol{p}'_3 \; \delta^3(\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 - \boldsymbol{P}_B) \delta^3(\boldsymbol{p}'_1 + \boldsymbol{p}'_2 + \boldsymbol{p}'_3 - \boldsymbol{P}'_B) \\ & \times \psi^*_{n_B L_B M_{L_B}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3) \psi_{n'_B L'_B M'_{L_B}}(\boldsymbol{p}'_1, \boldsymbol{p}'_2, \boldsymbol{p}'_3) \\ & \times \langle q_1(\boldsymbol{p}_1) q_2(\boldsymbol{p}_2) q_3(\boldsymbol{p}_3)\rangle \, q_1(\boldsymbol{p}'_1) q_2(\boldsymbol{p}'_2) q_3(\boldsymbol{p}'_3). \end{aligned} \tag{88}$$

Using the orthonormality of quark momentum states,

$$\langle q_1(\boldsymbol{p}_1) q_2(\boldsymbol{p}_2) q_3(\boldsymbol{p}_3)\rangle \, q_1(\boldsymbol{p}'_1) q_2(\boldsymbol{p}'_2) q_3(\boldsymbol{p}'_3) = \delta^3(\boldsymbol{p}_1 - \boldsymbol{p}'_1) \delta^3(\boldsymbol{p}_2 - \boldsymbol{p}'_2) \delta^3(\boldsymbol{p}_3 - \boldsymbol{p}'_3), \tag{89}$$

we simplify:

$$\langle B(\boldsymbol{P}_B)\rangle \, B'(\boldsymbol{P}'_B) = 2E_B \int d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 \; \delta^3(\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 - \boldsymbol{P}_B) \delta^3(\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 - \boldsymbol{P}'_B) \tag{90}$$

$$\times \psi^*_{n_B L_B M_{L_B}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3) \psi_{n'_B L'_B M'_{L_B}}(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3). \tag{91}$$

Now, we change the coordinate and volume element

$$d^3\boldsymbol{p}_1 d^3\boldsymbol{p}_2 d^3\boldsymbol{p}_3 = \frac{1}{3^{3/2}} d^3\boldsymbol{P} d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho \tag{92}$$

so that

$$\langle B(\boldsymbol{P}_B)\rangle\, B'(\boldsymbol{P}'_B) = 2E_B \int \frac{1}{3^{3/2}} d^3\boldsymbol{P} d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho\ \delta^3(\boldsymbol{P}-\boldsymbol{P}_B)\delta^3(\boldsymbol{P}-\boldsymbol{P}'_B)|\psi_{n_B L_B M_{L_B}}(\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3)|^2 \tag{93}$$

Now, integrating over $\boldsymbol{P}$ using the delta function:

$$d^3\boldsymbol{P}\ \delta^3(\boldsymbol{P}-\boldsymbol{P}_B)\delta^3(\boldsymbol{P}-\boldsymbol{P}'_B) = \delta^3(\boldsymbol{P}_B-\boldsymbol{P}'_B). \tag{94}$$

We obtain

$$\langle B(\boldsymbol{P}_B)\rangle\, B'(\boldsymbol{P}'_B) = 2E_B\delta^3(\boldsymbol{P}_B-\boldsymbol{P}'_B)\frac{1}{3^{3/2}} \int d^3\boldsymbol{p}_\lambda d^3\boldsymbol{p}_\rho\ |\psi_{n_B L_B M_{L_B}}(\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3)|^2 \tag{95}$$

where the wave function can be written as

$$\psi_{nlm}(\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3) = N\psi_{n_\lambda l_\lambda m_\lambda}(\boldsymbol{p}_\lambda)\psi_{n_\rho l_\rho m_\rho}(\boldsymbol{p}_\rho). \tag{96}$$

with some normalization N. For proper normalization,

$$\frac{N^2}{3^{3/2}} \int d^3\boldsymbol{p}_\lambda\ |\psi_{n_\lambda l_\lambda m_\lambda}(\boldsymbol{p}_\lambda)|^2 \int d^3\boldsymbol{p}_\rho\ |\psi_{n_\rho l_\rho m_\rho}(\boldsymbol{p}_\rho)|^2 = 1, \tag{97}$$

we obtain $N = 3^{3/4}$. For instance, let us assume S wave case, we have

$$\psi_{000}(\boldsymbol{p}_1,\boldsymbol{p}_2,\boldsymbol{p}_3) = \left(\frac{3}{(\pi\beta_\rho^2)(\pi\beta_\lambda^2)}\right)^{3/4} \exp\left(-\frac{\boldsymbol{p}_\rho^2}{2\beta_\rho^2}\right)\exp\left(-\frac{\boldsymbol{p}_\lambda^2}{2\beta_\lambda^2}\right). \tag{98}$$

In the end, we recover the final result

$$\langle B(\boldsymbol{P}_B)\rangle\, B'(\boldsymbol{P}'_B) = 2E_B\delta^3(\boldsymbol{P}_B-\boldsymbol{P}'_B). \tag{99}$$

E The $^3P_0$ Model and Quantum Number Constraints

- Total angular momentum: $J = 0$
- Parity: $P = +1$
- Charge conjugation: $C = +1$
- Spin: $S = 1$ (triplet state)
- Orbital angular momentum: $L = 1$ (P-wave)

Thus, the created pair is in a $^3P_0$ state, which ensures consistency with the vacuum quantum numbers $J^{PC} = 0^{++}$.

A singlet $^1S_0$ quark-antiquark pair has:

- Spin: $S = 0$
- Orbital angular momentum: $L = 0$
- Total angular momentum: $J = 0$ (since $J = L + S$)

To check if it matches the vacuum, we compute its parity and charge conjugation:

Parity: The parity of a $q\bar{q}$ pair is given by

$$P = (-1)^L(-1)^{S+1}. \tag{100}$$

For the $^1S_0$ state:

$$P = (-1)^0(-1)^{0+1} = (-1). \tag{101}$$

Since the vacuum has $P = +1$, the parity does not match, making this state forbidden.

Charge Conjugation: The charge conjugation quantum number is given by

$$C = (-1)^{L+S}. \tag{102}$$

For the $^1S_0$ state:

$$C = (-1)^{0+0} = (+1), \tag{103}$$

which matches the vacuum. However, since parity does not match, the singlet state $^1S_0$ is not allowed in the $^3P_0$ model.

For a triplet $^3P_0$ state:

- $S = 1$, $L = 1$, and $J = 0$.

Computing parity:

$$P = (-1)^1(-1)^{1+1} = (+1), \tag{104}$$

which matches the vacuum.

Computing charge conjugation:

$$C = (-1)^{1+1} = (+1), \tag{105}$$

which also matches the vacuum.

Since both P and C match the vacuum, the $^3P_0$ state is allowed in hadronic decays.

The $^3P_0$ model assumes that the decay of a hadron occurs via the creation of a quark-antiquark pair with vacuum quantum numbers $J^{PC} = 0^{++}$. The singlet $^1S_0$ state is forbidden because its parity does not match the vacuum, while the triplet $^3P_0$ state is allowed since it satisfies all quantum number constraints.

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